Vector Fields and Line Integrals

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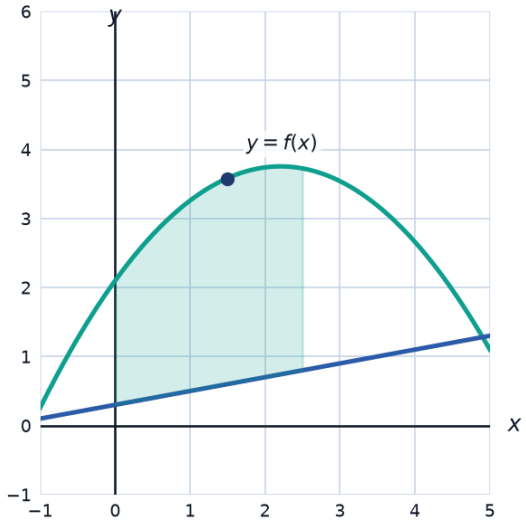
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Vector fields and divergence

For the vector field $\mathbf{F}(x, y) = \langle x^2 y,\ xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

2 [Foundational] • Target: Line integrals

Evaluate the line integral $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.



3 [Foundational] • Target: Line integrals

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \le t \le 1$.

4 [Foundational] • Target: Conservative vector fields and potential functions

Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

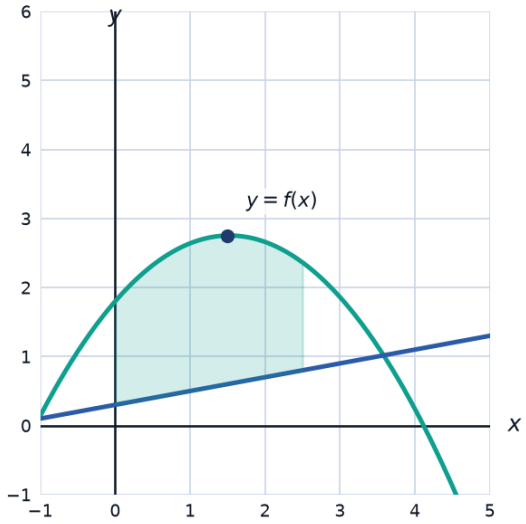
5 [Foundational] • Target: Conservative vector fields and potential functions

Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.

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6 [Foundational] • Target: The Fundamental Theorem for line integrals

Using the potential function $f(x, y) = x^2y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.



7 [Foundational] • Target: Green's Theorem

Use Green's Theorem, $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, to evaluate $\oint_C xy\,dx + x^2\,dy$ where C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

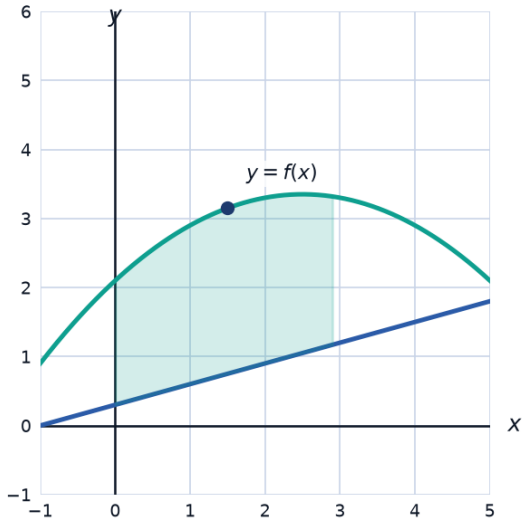
8 [Intermediate] • Target: Green's Theorem

Use Green's Theorem to find the area of the region enclosed by a curve C, using the formula $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$.

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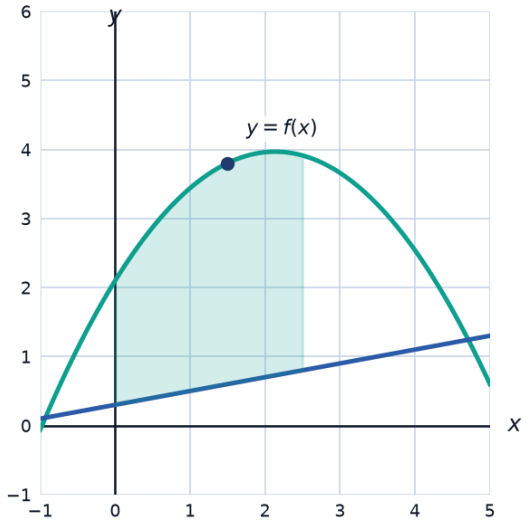
9 [Intermediate] • Target: Green's Theorem

Use Green's Theorem to evaluate $\oint_C (3y)\,dx + (5x)\,dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.



10 [Intermediate] • Target: Work done by a force field

A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.



11 [Intermediate] • Target: Work done by a force field

Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

12 [Intermediate] • Target: Testing for conservative fields

Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.

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13 [Intermediate] • Target: Testing for conservative fields

Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

14 [Intermediate] • Target: Applying Green's Theorem to circulation

Use Green's Theorem to evaluate the circulation $\oint_C (-y)\,dx + x\,dy$ around the boundary of the rectangle $0 \le x \le 3$, $0 \le y \le 2$, traversed counterclockwise.

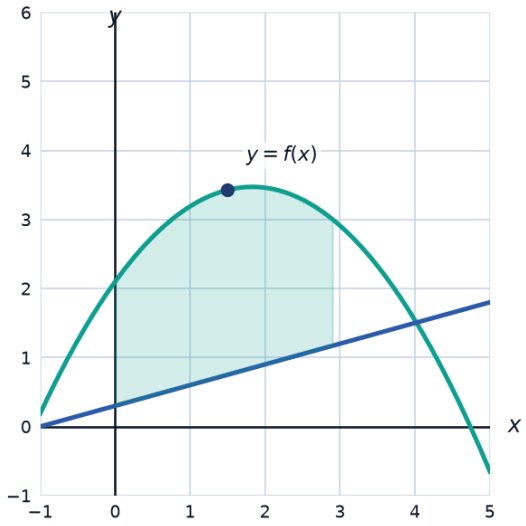
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 9.01 Separable Differential Equations

Evaluate $\int_C y\,dx + x\,dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.

16 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 1.01 Precalculus Review for Calculus

Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.



17 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n}\,ds = \iint_D \nabla \cdot \mathbf{F}\,dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \le 4$.

18 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 11.01 Sequences and Series

A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.

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19 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.01 Convergence Tests

For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:

a Find the divergence $\nabla \cdot \mathbf{F}$.

b Find the curl $\nabla \times \mathbf{F}$.

c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D.

20 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector fields and divergence

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$

c. 4π

d. 12π

22 [Foundational] • Target: Line integrals

Work done by $\mathbf{F} = (y, x, 0)$ along line from $(0, 0, 0)$ to $(1, 1, 0)$:

a. 0

b. 2

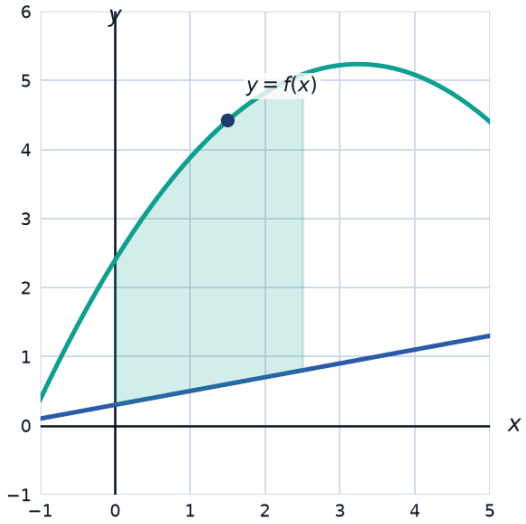
c. 1

d. 1/2

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23 [Foundational] • Target: Conservative vector fields and potential functions

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2+y^2+z^2)dV =$



a. $4\pi/3$

b. 3π

c. $12\pi/5$

d. π

24 [Foundational] • Target: The Fundamental Theorem for line integrals

Green's theorem in the form $\iint_D (\partial Q/\partial x - \partial P/\partial y)dA = \oint_C P dx + Q dy$. For $P = 0, Q = x$: $\oint_C x\, dy =$

a. $-\iint_D dA$

b. 0

c. $\iint_D dA = \text{Area}(D)$

d. $\oint_C y\, dx$

25 [Foundational] • Target: Green's Theorem

A conservative vector field $\mathbf{F} = ablaf$ has $abla \times \mathbf{F} =$

a. $\mathbf{F}$

b. f

c. $ablaf$

d. $\mathbf{0}$

26 [Foundational] • Target: Work done by a force field

Line integral of scalar field: $\int_C f\, ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\, ds =$

a. area

b. volume

c. 0

d. arc length

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27 [Foundational] • Target: Testing for conservative fields

Normal line to $z = x^2 + y^2$ at $(1, 1, 2)$: direction:

a. $(2, 2, -1)$

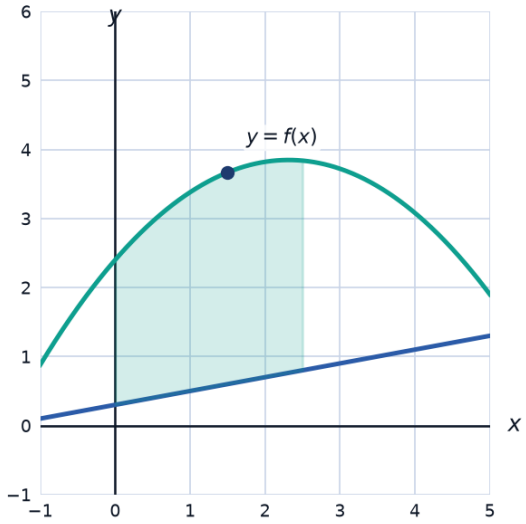
b. $(1, 1, 1)$

c. $(-2, -2, 1)$

d. $(1, 0, -2)$

28 [Foundational] • Target: Applying Green's Theorem to circulation

Find the flux of $\mathbf{F} = \langle x, y, z \rangle$ through the closed unit sphere using the Divergence Theorem.



a. 2π

b. $\frac{4\pi}{3}$

c. 4π

d. $\frac{8\pi}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Line integrals over piecewise paths

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 2$.

a. 32π

b. 8π

c. -16π

d. 16π

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30 **[Intermediate]** • **Target: Flux and Green's Theorem (normal form)**

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the cube $[0, 1]^3$.

a. 3

b. 6

c. $\frac{3}{2}$

d. -3

31 **[Intermediate]** • **Target: A capstone conservative-field application**

Stokes' theorem for $\mathbf{F} = (y, -x, 0)$ over unit disk $z = 0$: $\oint \mathbf{F} \cdot d\mathbf{r} =$

a. 0

b. 2π

c. -2π

d. π

32 **[Intermediate]** • **Target: Vector fields and divergence**

Flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, dA$. For $\mathbf{F} = (0, 0, 1)$ over upward-facing plane $z = 0$, $D = \{x^2 + y^2 \le 1\}$:

a. 0

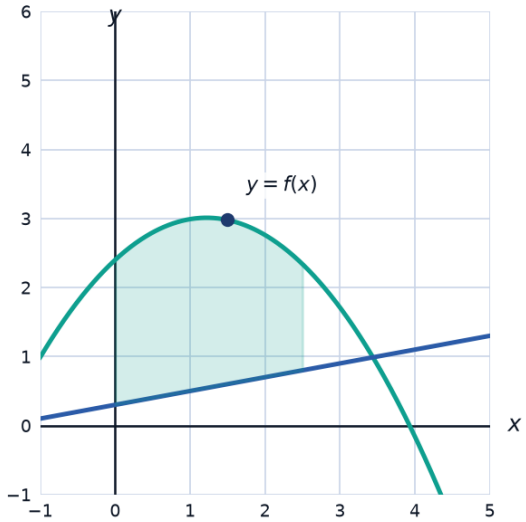
b. 1

c. 2π

d. π

33 **[Intermediate]** • **Target: Line integrals**

Green's theorem: $\oint_C y^2\, dx + 2xy\, dy$ over boundary of $[0, 2] \times [0, 3]$:



a. 6

b. 0

c. 12

d. 3

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34 **[Intermediate]** • **Target: Conservative vector fields and potential functions**

Gradient theorem (FTC for line integrals): $\int_C abla f \cdot d\mathbf{r} =$

a. $\iint f\,dA$

b. $f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

c. $\oint f\,ds$

d. $abla^2 f$

35 **[Intermediate]** • **Target: The Fundamental Theorem for line integrals**

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the upper hemisphere $x^2 + y^2 + z^2 = 1,\ z \geq 0$, with outward normal.

a. $\frac{4\pi}{3}$

b. $-\frac{2\pi}{3}$

c. $\frac{2\pi}{3}$

d. $\frac{\pi}{3}$

36 **[Intermediate]** • **Target: Green's Theorem**

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle y,\ -x, xz \rangle$ around the circle $x^2 + y^2 = 9$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. -18π

b. 18π

c. -9π

d. -36π

37 **[Intermediate]** • **Target: Work done by a force field**

Evaluate $\oint_C (y\,dx - x\,dy)$ around the circle $x^2 + y^2 = 4$ (counterclockwise) using Green's Theorem.

a. 4π

b. 8π

c. -4π

d. -8π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Testing for conservative fields • Cross-Lesson: 11.01 Sequences and Series

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 1, 1, 1 \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

a. $\frac{27}{2}$

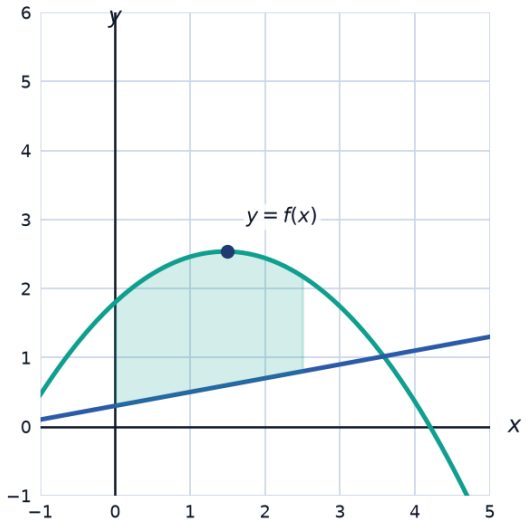
b. 27

c. $-\frac{27}{2}$

d. $\frac{27}{4}$

39 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 12.01 Convergence Tests

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$



a. $f(c)(b - a)$

b. $f'(c)(b - a)$

c. $f(b) - f(a)$

d. $f(c)$

40 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 12.02 Power Series and Taylor Series

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x + h) - F(x)$ as:

a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x + h]$

c. $f(x) \cdot (x + h)$

d. f(x+h)-f(x)

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41 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Where does the line $\mathbf{r}(t) = \langle 3, -3, -1\rangle + t\langle 0, 0, 1\rangle$ meet the plane $3x + y + 3z = 4$?

a. $\left(3, -3, \frac{1}{3}\right)$

b. $\left(3, -3, -\frac{2}{3}\right)$

c. $\left(3, -3, -\frac{4}{3}\right)$

d. $(3, -3, -1)$

42 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Line integral: $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}'(t)\,dt$. For $\mathbf{F} = \mathbf{r}$, C line from 0 to $(1, 0, 0)$:

a. 0

b. 1

c. 1/2

d. 2

43 [Challenging] • Target: Vector fields and divergence • Cross-Lesson: 15.01 Double Integrals

Fubini's theorem allows interchanging iterated integrals when:

a. both integrals converge

b. f is continuous

c. $\iint |f| < \infty$

d. $f \geq 0$

44 [Challenging] • Target: Line integrals • Cross-Lesson: 2.01 Evaluating Limits

Which theorem converts a surface integral of $abla \times \mathbf{F}$ to a line integral?

a. Green's Theorem

b. Stokes' Theorem

c. Gauss's Divergence Theorem

d. Cauchy's Integral Theorem

45 [Challenging] • Target: Conservative vector fields and potential functions • Cross-Lesson: 2.02 Continuity

Use Green's theorem to evaluate $\oint_C x - y\,dx + x + y\,dy$, where C is the boundary of the rectangle $[0, 1] \times [0, 3]$, oriented counterclockwise.

a. 6

b. -6

c. 12

d. 3

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